

KUPINO, Russian Federation

WMO: 297060

Lat: 54.3869N

Lon: 77.3083E

Elev: 381

StdP: 14.49

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.6	-26.0	-37.2	0.7	-31.4	-32.0	0.9	-25.9	27.8	12.7	23.4	16.9	5.3	250	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	87.0	65.4	83.9	64.5	80.9	63.4	68.9	79.9	67.3	78.1	65.9	76.6	9.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	94.7	72.8	63.6	89.1	71.8	61.9	83.9	70.6	33.3	79.4	32.0	78.3	30.9	76.4	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.7	18.3	DB	-35.3	92.8	8.1	3.7	-41.1	95.5	-45.8	97.6	-50.3	99.7	-56.2	102.3	(4)
(5)			WB	-35.8	71.8	7.9	2.4	-41.5	73.5	-46.1	75.0	-50.6	76.3	-56.4	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.8	-1.2	3.1	19.6	40.6	54.3	64.3	67.3	62.8	52.0	38.3	20.8	5.6	(6)
(7)		DBStd	26.43	14.64	13.15	10.84	10.72	9.66	7.53	6.01	7.00	8.53	9.27	13.01	14.40	(7)
(8)		HDD50	6941	1588	1313	943	317	70	4	0	1	75	377	876	1377	(8)
(9)		HDD65	10943	2053	1733	1408	731	350	100	47	126	399	828	1326	1842	(9)
(10)		CDD50	1757	0	0	0	36	202	435	535	399	136	14	0	0	(10)
(11)		CDD65	283	0	0	0	0	17	81	117	59	9	0	0	0	(11)
(12)		CDH74	2974	0	0	0	14	303	821	1038	663	135	0	0	0	(12)
(13)		CDH80	825	0	0	0	1	70	230	296	196	32	0	0	0	(13)
(14)	Wind	WSAvg	8.8	8.9	9.3	10.1	9.7	9.5	7.7	7.1	7.3	7.9	8.7	9.6	9.6	(14)
(15)	Precipitation	PrecAvg	12.2	0.5	0.4	0.4	0.6	1.0	1.5	2.1	2.1	1.2	1.0	0.7	0.6	(15)
(16)		PrecMax	18.3	1.5	1.1	1.3	1.9	2.7	4.0	5.0	5.7	3.4	2.5	2.1	1.4	(16)
(17)		PrecMin	7.3	0.1	0.0	0.0	0.0	0.2	0.2	0.0	0.2	0.0	0.1	0.0	0.1	(17)
(18)		PrecStd	2.4	0.3	0.2	0.2	0.5	0.6	1.0	1.3	1.3	0.7	0.6	0.5	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	33.5	44.0	75.9	86.9	89.9	91.3	90.9	84.3	68.6	46.7	34.3	(19)
(20)			MCWB	32.2	32.0	38.1	55.4	61.5	64.9	68.2	66.3	60.4	51.9	42.3	32.6	(20)
(21)		2%	DB	28.5	30.5	38.8	68.9	81.5	85.8	86.5	85.3	77.7	62.2	42.9	32.3	(21)
(22)			MCWB	27.2	29.3	35.4	53.0	59.6	64.8	66.9	65.2	58.7	49.6	39.6	31.4	(22)
(23)		5%	DB	24.6	26.8	36.2	63.2	77.5	82.3	83.3	81.0	71.7	56.5	39.0	29.1	(23)
(24)			MCWB	23.5	25.5	33.7	49.9	57.5	64.2	65.9	63.2	56.3	47.2	36.6	27.9	(24)
(25)		10%	DB	20.8	22.8	34.1	57.8	72.3	78.7	80.0	77.2	66.7	51.5	35.7	25.4	(25)
(26)			MCWB	19.8	21.5	32.1	47.4	55.4	62.7	65.2	62.4	54.0	44.6	33.6	24.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	32.2	38.6	57.5	64.7	70.1	71.6	69.5	63.9	53.8	43.9	33.1	(27)
(28)			MCD B	33.3	33.3	42.7	71.9	82.0	82.8	84.3	81.0	79.6	63.1	45.3	33.8	(28)
(29)		2%	WB	27.3	29.2	35.8	54.4	61.1	67.4	69.6	67.2	60.1	51.0	39.9	31.5	(29)
(30)			MCD B	28.3	30.4	38.4	66.4	76.8	79.2	80.3	79.4	73.3	59.9	42.4	32.4	(30)
(31)		5%	WB	23.6	25.7	34.0	50.7	58.8	65.8	68.1	65.5	57.6	48.0	36.9	28.0	(31)
(32)			MCD B	24.5	26.8	35.9	61.7	73.9	77.5	78.0	77.1	70.0	55.1	39.1	29.1	(32)
(33)		10%	WB	19.8	21.5	32.1	47.7	56.6	64.2	66.7	63.7	55.1	45.3	33.7	24.3	(33)
(34)			MCD B	20.8	22.7	34.0	57.4	69.7	75.3	76.5	74.9	65.0	51.1	35.4	25.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.2	15.0	14.9	18.1	21.7	19.9	19.0	20.3	19.1	15.2	11.8	13.8	(35)
(36)			MCDBR	15.2	13.7	12.6	27.0	29.0	24.9	24.4	26.9	28.1	23.7	11.4	14.3	(36)
(37)			MCWBR	15.2	13.5	10.6	14.2	12.5	10.1	9.3	10.4	12.9	13.2	9.4	14.2	(37)
(38)		5% WB	MCD BR	15.3	13.7	11.9	25.1	25.2	21.1	19.9	22.3	24.8	21.3	11.0	14.0	(38)
(39)			MCWBR	15.3	13.6	10.5	13.9	12.7	10.0	8.8	10.2	12.7	12.8	9.5	14.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.321	0.379	0.403	0.404	0.411	0.435	0.421	0.368	0.351	0.325	0.292	(40)	
(41)		taud	2.033	1.993	1.932	2.157	2.224	2.291	2.223	2.228	2.330	2.250	2.094	1.990	(41)	
(42)		Ebn at Noon	195	231	243	259	266	266	256	252	251	223	179	161	(42)	
(43)		Edh at Noon	23	34	45	42	41	39	41	39	30	26	22	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	334	682	1189	1492	1816	1937	1767	1463	1045	535	252	218	(44)	
(45)		RadStd	23	30	80	112	148	140	118	121	101	69	34	24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page